Off-diagonal Commonality of Cycles of Length $2k$ and $2k+1$

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August 20, 2026

Abstract

For an integer $k \geq 2$, we determine exactly the parameters for which (C_{2k}, C_{2k+1}) is a common pair of graphs in the sense of Behague, Morrison, and Noel. Let $a_k^\star \in (1/2, 1)$ be the unique solution of

$$\frac{(a_k^\star)^{2k-1}(a_k^\star + 2k)}{2k+1} = 2^{1-2k}.$$

We prove

$$\pi(C_{2k}, C_{2k+1}) = (0, a_k^\star].$$

The upper bound is witnessed by the balanced disjoint union of two cliques. For the lower bound, after passing to a finite-rank graphon operator, a rank-one case of Lidskii's majorisation theorem, combined with Karamata's inequality, separates the Perron eigenvalue from the remaining spectrum. A Hilbert–Schmidt bound shows that at most one negative eigenvalue can contribute with the wrong sign. Perron–Frobenius-type domination then compensates for that eigenvalue exactly, reducing the graphon inequality to the elementary convexity inequality $x^{2k} + (1-x)^{2k} \geq 2^{1-2k}$.

1 Definitions and statement of the result

A *graphon* is a symmetric measurable function $W\colon [0,1]^2 \to [0,1]$. For a finite graph H, its homomorphism density in W is

$$t(H, W) := \int_{[0,1]^{V(H)}} \prod_{uv \in E(H)} W(x_u, x_v) \prod_{u \in V(H)} dx_u.$$

Write $e(H) = |E(H)|$. For background on graphons and homomorphism densities we refer to Lovász [3].

Following Behague, Morrison, and Noel [1, Definition 1.1], for $a, b \in (0,1)$ with $a + b = 1$, a pair (H_1, H_2) is said to be (a,b)-*common* if

$$\frac{t(H_1, W)}{e(H_1)a^{e(H_1)-1}} + \frac{t(H_2, 1-W)}{e(H_2)b^{e(H_2)-1}} \geq \frac{a}{e(H_1)} + \frac{b}{e(H_2)} \tag{1}$$

for every graphon W. Define

$$\pi(H_1, H_2) := \{a \in (0,1) : (H_1, H_2) \text{ is } (a, 1-a)\text{-common}\}.$$

Theorem 1.1 (Exact adjacent-cycle commonality region). *Let $k \geq 2$. There is a unique number $a_k^\star \in (1/2, 1)$ satisfying*

$$\frac{(a_k^\star)^{2k-1}(a_k^\star + 2k)}{2k+1} = 2^{1-2k}. \tag{2}$$

Then

$$\pi(C_{2k}, C_{2k+1}) = (0, a_k^\star]. \tag{3}$$

Equivalently, for $a \in (0,1)$ and $b = 1-a$, the inequality

$$\frac{t(C_{2k}, W)}{2k\, a^{2k-1}} + \frac{t(C_{2k+1}, 1-W)}{(2k+1)b^{2k}} \geq \frac{a}{2k} + \frac{b}{2k+1} \tag{4}$$

holds for every graphon W if and only if $a \leq a_k^\star$.

The critical equation may also be written as

$$(2a_k^\star)^{2k-1}(2a_k^\star + 4k) = 4k + 2. \tag{5}$$

For $k = 2$, this becomes

$$8(a_2^\star)^3(a_2^\star + 4) = 5,$$

so Theorem 1.1 includes the exact determination of $\pi(C_4, C_5)$. Numerically, the first few critical values are

k	$a_k^\star$
2	0.5172134403...
3	0.5073512819...
4	0.5040650275...
5	0.5025774302...
6	0.5017794484...
7	0.5013021447...

and $a_k^\star \to 1/2$ as $k \to \infty$, by the bound (26) below.

2 Proof of the main theorem

We first reduce the proof to step graphons. The details are included so that no compactness theorem for graphons is needed.

Lemma 2.1 (Step-graphon reduction)**.** *Fix integers $r_1, r_2 \geq 3$ and a continuous function $\Phi\colon \mathbb{R}^2 \to \mathbb{R}$. Suppose that*

$$\Phi\big(t(C_{r_1}, W), t(C_{r_2}, 1-W)\big) \geq 0$$

for every symmetric step graphon W. Then the same inequality holds for every graphon W.

Proof. For $m \geq 1$, partition $[0,1]$ into the 2^m dyadic intervals of length 2^{-m}, and let $\mathcal{A}_m$ be the finite sigma-algebra generated by this partition. Set

$$W_m := \mathbb{E}[W \mid \mathcal{A}_m \otimes \mathcal{A}_m].$$

Conditional expectation preserves the bounds $0 \leq W \leq 1$, and symmetry is preserved because $\mathcal{A}_m \otimes \mathcal{A}_m$ is invariant under interchanging the two coordinates. Thus W_m is a symmetric step graphon.

The spaces of dyadic rectangle step functions are nested and their union is dense in $L^2([0,1]^2)$. Given $\varepsilon > 0$, choose such a step function G, measurable with respect to $\mathcal{A}_M \otimes \mathcal{A}_M$, for which $\|W - G\|_2 < \varepsilon$. For $m \geq M$, the L^2-contraction property of conditional expectation gives

$$\|W_m - G\|_2 = \|\mathbb{E}[W - G \mid \mathcal{A}_m \otimes \mathcal{A}_m]\|_2 \leq \|W - G\|_2 < \varepsilon.$$

Hence $\|W_m - W\|_2 < 2\varepsilon$ for all $m \geq M$, so $\|W_m - W\|_2 \to 0$. In particular, $\|W_m - W\|_1 \to 0$. Also

$$1 - W_m = \mathbb{E}[1 - W \mid \mathcal{A}_m \otimes \mathcal{A}_m],$$

so $\|(1-W_m)-(1-W)\|_1 \to 0$.

We claim that for every fixed integer $r \geq 3$,

$$|t(C_r, W_m) - t(C_r, W)| \leq r\|W_m - W\|_1. \tag{6}$$

Indeed, write $x_{r+1} = x_1$ and use the telescoping identity

$$\prod_{i=1}^{r} a_i - \prod_{i=1}^{r} b_i = \sum_{j=1}^{r} \left(\prod_{i<j} a_i \right) (a_j - b_j) \left(\prod_{i>j} b_i \right).$$

Taking

$$a_i = W_m(x_i, x_{i+1}), \qquad b_i = W(x_i, x_{i+1}),$$

and using $0 \leq a_i, b_i \leq 1$, integration gives (6). The same argument applies to $1-W_m$ and $1-W$. Thus both cycle densities in the asserted inequality converge. Continuity of Φ now lets the step-graphon inequality pass to the limit. □

Now let K be a symmetric step kernel, constant on every rectangle $B_i \times B_j$ of a finite measurable partition

$$[0,1] = B_1 \sqcup \cdots \sqcup B_N, \qquad |B_i| > 0.$$

Let T_K be the associated integral operator,

$$(T_K f)(x) = \int_0^1 K(x,y) f(y)\, dy.$$

Put

$$E := \operatorname{span}\{\mathbf{1}_{B_1}, \ldots, \mathbf{1}_{B_N}\}.$$

Then T_K maps $L^2([0,1])$ into E and vanishes on $E^\perp$. In the orthonormal basis

$$e_i := \frac{\mathbf{1}_{B_i}}{\sqrt{|B_i|}},$$

the restriction $T_K|_E$ is represented by the real symmetric matrix

$$\big(K_{ij}\sqrt{|B_i||B_j|}\big)_{i,j=1}^{N},$$

where K_{ij} is the value of K on $B_i \times B_j$.

Lemma 2.2 (Cycle densities are traces)**.** *For every integer $r \geq 3$,*

$$t(C_r, K) = \operatorname{Tr}(T_K^r), \tag{7}$$

where the trace is taken on the finite-dimensional space E.

Proof. Expanding the matrix trace in the basis $(e_i)_{i=1}^N$ gives

$$\operatorname{Tr}(T_K^r) = \sum_{i_1,\ldots,i_r=1}^{N} |B_{i_1}| \cdots |B_{i_r}| K_{i_1 i_2} K_{i_2 i_3} \cdots K_{i_r i_1},$$

which is exactly the step-function expansion of $t(C_r, K)$. □

Let $J := T_1$ be the operator of the constant kernel 1. Since the underlying measure space has total mass one,

$$Jf = \langle \mathbf{1}, f\rangle \mathbf{1},$$

so J is the rank-one orthogonal projection onto $\mathbb{R}\mathbf{1}$. If $U = 1 - W$ and $T := T_U$, then

$$T_W = J - T. \tag{8}$$

For a vector $x \in \mathbb{R}^N$, write $x^{\downarrow}$ for its coordinates rearranged in nonincreasing order. Recall that x is *majorised* by y, written $x \prec y$, if

$$\sum_{i=1}^{j} x_i^{\downarrow} \le \sum_{i=1}^{j} y_i^{\downarrow} \quad (1 \le j < N), \qquad \sum_{i=1}^{N} x_i = \sum_{i=1}^{N} y_i.$$

Karamata's inequality (see, e.g., [4, Chapter 4]) states that if $x \prec y$ and $\varphi\colon \mathbb{R} \to \mathbb{R}$ is convex, then

$$\sum_{i=1}^{N} \varphi(x_i) \le \sum_{i=1}^{N} \varphi(y_i). \tag{9}$$

The next lemma is the special rank-one form of Lidskii's additive majorisation theorem that we need; for the general form see [2, Chapter III] or [4, Chapter 9]. We include a self-contained proof.

Lemma 2.3 (Rank-one lower majorisation)**.** *Let A be an $N \times N$ real symmetric matrix with eigenvalues*

$$\alpha_1 \ge \alpha_2 \ge \cdots \ge \alpha_N,$$

and let P be a rank-one orthogonal projection. Let

$$\mu_1 \ge \mu_2 \ge \cdots \ge \mu_N$$

be the eigenvalues of $A + P$. Then

$$(\alpha_1, \ldots, \alpha_{N-1}, \alpha_N + 1) \prec (\mu_1, \ldots, \mu_N). \tag{10}$$

Proof. The eigenvalues of a positive rank-one perturbation interlace:

$$\mu_i \ge \alpha_i \ge \mu_{i+1} \qquad (1 \le i < N), \tag{11}$$

and also $\mu_N \ge \alpha_N$. This follows directly from the Courant–Fischer min–max principle; equivalently, it is the rank-one case of Weyl interlacing (see, e.g., [2, Chapter III]).

Set

$$v := (\alpha_1, \ldots, \alpha_{N-1}, \alpha_N + 1).$$

Fix $1 \le j < N$. The sum of the j largest coordinates of v is the maximum of $\sum_{i\in S} v_i$ over all j-element subsets $S \subseteq \{1, \ldots, N\}$.

If $N \notin S$, then

$$\sum_{i\in S} v_i \le \sum_{i=1}^{j} \alpha_i \le \sum_{i=1}^{j} \mu_i,$$

by the first inequality in (11).

If $N \in S$, then

$$\sum_{i\in S} v_i \le 1 + \alpha_N + \sum_{i=1}^{j-1} \alpha_i. \tag{12}$$

Since $\mathrm{Tr}(A+P) = \mathrm{Tr}(A) + 1$ and $\mu_{i+1} \le \alpha_i$ for $j \le i < N$, we have

$$\begin{aligned}\sum_{i=1}^{j} \mu_i &= \mathrm{Tr}(A+P) - \sum_{i=j+1}^{N} \mu_i \\ &\ge 1 + \sum_{i=1}^{N} \alpha_i - \sum_{i=j}^{N-1} \alpha_i \\ &= 1 + \alpha_N + \sum_{i=1}^{j-1} \alpha_i.\end{aligned}$$

This dominates the right side of (12). Thus the first j majorisation inequality holds. Equality of the total sums follows from $\mathrm{Tr}(A+P) = \mathrm{Tr}(A) + 1$. $\square$

Corollary 2.4 (Even trace under subtraction of a rank-one projection)**.** *Let T be a real symmetric $N \times N$ matrix with eigenvalues*

$$\lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_N,$$

and let P be a rank-one orthogonal projection. If $n \ge 2$ is even, then

$$\mathrm{Tr}(P-T)^n \ge (1-\lambda_1)^n + \sum_{i=2}^{N} |\lambda_i|^n. \tag{13}$$

Proof. Apply Lemma 2.3 to $A = -T$. The nonincreasingly ordered eigenvalues of $-T$ are

$$-\lambda_N \ge \cdots \ge -\lambda_1.$$

Hence

$$(-\lambda_N, \ldots, -\lambda_2, 1-\lambda_1) \prec \lambda(P-T),$$

where $\lambda(P-T)$ denotes the eigenvalue vector of $P-T$. Apply Karamata's inequality to the convex function $x \mapsto x^n$. Since n is even, $(-\lambda_i)^n = |\lambda_i|^n$, and (13) follows. $\square$

The following elementary bounds use the pointwise condition $0 \le U \le 1$. They do not assume that T_U is positive semidefinite.

Lemma 2.5 (Perron–Frobenius-type domination and the tail sum bound)**.** *Let U be a symmetric step graphon, choose a finite partition on which U is constant, let E be its associated step-function space, and let $T = T_U|_E$. Let*

$$\lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_N$$

be the eigenvalues of T. Then

$$0 \le \lambda_1 \le 1, \tag{14}$$

$$|\lambda_i| \le \lambda_1 \quad (1 \le i \le N), \tag{15}$$

$$\sum_{i=2}^{N} \lambda_i^2 \le \lambda_1(1-\lambda_1) \le \frac{1}{4}. \tag{16}$$

Proof. Let f be a real unit vector in E. Since the elements of E are constant on every cell of the partition, $|f|$ is again a unit vector in E. Pointwise nonnegativity of U gives

$$|\langle f, Tf\rangle| \le \langle |f|, T|f|\rangle, \tag{17}$$

and by the Rayleigh principle the right side is at most λ_1. Applying (17) to a unit eigenvector of λ_i shows that $|\lambda_i| \le \lambda_1$, proving (15).

Next, the pointwise bound $U \le 1$ gives

$$\langle |f|, T|f| \rangle \le \left(\int_0^1 |f| \right)^2,$$

and the Cauchy–Schwarz inequality on the probability space $[0,1]$ gives $\int_0^1 |f| \le \|f\|_2 = 1$. Hence $\lambda_1 \le 1$. Also $\lambda_1 \ge 0$, for example by testing the nonnegative unit vector $\mathbf{1}$. This proves (14).

Finally, put

$$s := \int_{[0,1]^2} U(x,y)\, dx\, dy = \langle \mathbf{1}, T\mathbf{1} \rangle.$$

Since $\|\mathbf{1}\|_2 = 1$, the Rayleigh principle gives $s \le \lambda_1$. On the other hand,

$$\sum_{i=1}^{N} \lambda_i^2 = \mathrm{Tr}(T^2) = \int_{[0,1]^2} U(x,y)^2\, dx\, dy \le s.$$

Hence

$$\sum_{i=2}^{N} \lambda_i^2 \le s - \lambda_1^2 \le \lambda_1 - \lambda_1^2 \le \frac{1}{4},$$

as claimed. □

Fix an even integer

$$n \ge 4.$$

In the end we take $n = 2k$. For $a \in (0,1)$, put

$$b := 1 - a,$$

and define

$$\kappa = \kappa_n(a) := \frac{n a^{n-1}}{(n+1) b^n}, \qquad \rho_n(a) := \frac{a^{n-1}(a+n)}{n+1}. \tag{18}$$

Multiplying (4), with $n = 2k$, by na^{n-1} shows that the desired commonality inequality is equivalent to

$$t(C_n, W) + \kappa\, t(C_{n+1}, 1 - W) \ge \rho_n(a). \tag{19}$$

Indeed,

$$na^{n-1}\left(\frac{a}{n} + \frac{b}{n+1} \right) = a^n + \frac{n}{n+1} a^{n-1} b = \frac{a^{n-1}(a+n)}{n+1}.$$

Define

$$f_\kappa(x) := (1-x)^n + \kappa x^{n+1}, \qquad 0 \le x \le 1. \tag{20}$$

Then

$$\begin{aligned} f_\kappa'(x) &= -n(1-x)^{n-1} + \kappa(n+1)x^n, \\ f_\kappa''(x) &= n(n-1)(1-x)^{n-2} + \kappa n(n+1)x^{n-1} > 0. \end{aligned}$$

By the definition of κ,

$$f_\kappa'(b) = -na^{n-1} + \kappa(n+1)b^n = 0.$$

Therefore f_κ is strictly convex and has its unique minimum at b. Moreover,

$$\min_{0 \le x \le 1} f_\kappa(x) = f_\kappa(b) = a^n + \kappa b^{n+1} = \rho_n(a). \tag{21}$$

Lemma 2.6 (The critical point)**.** *There is a unique $\alpha_n^\star \in (1/2, 1)$ such that*

$$\rho_n(\alpha_n^\star) = 2^{1-n}. \tag{22}$$

Proof. Differentiation gives

$$\rho_n'(a) = \frac{na^{n-2}(a+n-1)}{n+1} > 0.$$

Also

$$\rho_n\left(\frac{1}{2}\right) = 2^{1-n}\frac{n+1/2}{n+1} < 2^{1-n}, \qquad \rho_n(1) = 1 > 2^{1-n}.$$

The intermediate value theorem and strict monotonicity give the result. $\square$

The next quantitative bounds are the only estimates on the critical point needed in the proof.

Lemma 2.7 (Bounds on the normalised coefficient)**.** *Let $\alpha_n^\star$ be as in Lemma 2.6, and put*

$$b_n^\star := 1-\alpha_n^\star, \qquad \kappa_n^\star := \kappa_n(\alpha_n^\star).$$

Then

$$\kappa_n^\star b_n^\star < 1, \qquad \kappa_n^\star < \frac{27}{13} < 2\sqrt{2}. \tag{23}$$

Consequently, for every $0 < a \le \alpha_n^\star$, with $b = 1-a$ and $\kappa = \kappa_n(a)$,

$$\kappa b < 1, \qquad \kappa < 2\sqrt{2}. \tag{24}$$

Proof. Write

$$\alpha_n^\star = \frac{1+\delta}{2}, \qquad b_n^\star = \frac{1-\delta}{2}, \qquad \delta \in (0,1).$$

Equation (22) is equivalent to

$$(1+\delta)^{n-1}(2n+1+\delta) = 2n+2. \tag{25}$$

Therefore

$$(1+\delta)^{n-1} < \frac{2n+2}{2n+1} = 1 + \frac{1}{2n+1}.$$

Bernoulli's inequality gives

$$1+(n-1)\delta \le (1+\delta)^{n-1},$$

and hence

$$0 < \delta < \frac{1}{(n-1)(2n+1)} \le \frac{1}{27}, \tag{26}$$

because $n \ge 4$.

Using (25), we compute

$$\begin{aligned}\kappa_n^\star b_n^\star &= \frac{n}{n+1}\left(\frac{1+\delta}{1-\delta}\right)^{n-1} \\ &= \frac{2n}{(2n+1+\delta)(1-\delta)^{n-1}}.\end{aligned} \tag{27}$$

Again by Bernoulli's inequality and (26),

$$(1-\delta)^{n-1} \ge 1-(n-1)\delta > 1 - \frac{1}{2n+1} = \frac{2n}{2n+1}.$$

Thus the denominator in (27) is strictly larger than $2n$, proving $\kappa_n^\star b_n^\star < 1$.

It follows that

$$\frac{1}{\kappa_n^\star} > b_n^\star = \frac{1-\delta}{2} > \frac{13}{27},$$

where the last inequality uses (26). Hence

$$\kappa_n^\star < \frac{27}{13} < 2\sqrt{2}.$$

Finally,

$$\kappa_n(a) = \frac{n}{n+1}\frac{a^{n-1}}{(1-a)^n}$$

is strictly increasing in a, and so is

$$\kappa_n(a)(1-a) = \frac{n}{n+1}\left(\frac{a}{1-a}\right)^{n-1}.$$

Therefore the endpoint bounds remain valid for every $a \le \alpha_n^\star$. □

We now prove that (19) holds whenever $a \le \alpha_n^\star$.

Proposition 2.8 (Spectral reduction). *Let $n \ge 4$ be even, let $\kappa > 0$, and let W be a symmetric step graphon. Choose a finite partition on which W is constant, let E be its associated step-function space, put $U = 1 - W$, and let $T = T_U|_E$. Order the eigenvalues of T as*

$$\lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_N.$$

Then

$$t(C_n, W) + \kappa t(C_{n+1}, U) \ge f_\kappa(\lambda_1) + \sum_{i=2}^{N} \lambda_i^n (1 + \kappa\lambda_i). \tag{28}$$

Proof. Let $P := J|_E$. Since $\mathbf{1} \in E$ and $\|\mathbf{1}\|_2 = 1$, the operator P is a rank-one orthogonal projection on E. Moreover, $T_W|_E = P - T$ by (8). Therefore, by Lemma 2.2 and Corollary 2.4,

$$t(C_n, W) = \mathrm{Tr}(P - T)^n \ge (1 - \lambda_1)^n + \sum_{i=2}^{N} |\lambda_i|^n.$$

Since n is even, $|\lambda_i|^n = \lambda_i^n$. Also

$$t(C_{n+1}, U) = \mathrm{Tr}(T^{n+1}) = \sum_{i=1}^{N} \lambda_i^{n+1}.$$

Adding the two expressions gives

$$\begin{aligned} t(C_n, W) + \kappa t(C_{n+1}, U) \ge (1 - \lambda_1)^n + \kappa\lambda_1^{n+1} \\ + \sum_{i=2}^{N} \big(\lambda_i^n + \kappa\lambda_i^{n+1}\big), \end{aligned}$$

which is exactly (28). □

Proposition 2.9 (Commonality up to the critical point). *Let $n \ge 4$ be even and $0 < a \le \alpha_n^\star$. Put $b = 1 - a$ and let κ be as in (18). Then every graphon W satisfies*

$$t(C_n, W) + \kappa t(C_{n+1}, 1 - W) \ge \rho_n(a). \tag{29}$$

Proof. Apply Lemma 2.1 with $\Phi(x,y) = x + \kappa y - \rho_n(a)$. It is enough to prove the assertion for a symmetric step graphon. Choose a finite partition on which W is constant, let E be its associated step-function space, set $U = 1 - W$, and let $T = T_U|_E$. Write its eigenvalues as

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N.$$

By Proposition 2.8,

$$t(C_n, W) + \kappa t(C_{n+1}, U) \geq f_\kappa(\lambda_1) + \sum_{i=2}^{N} \lambda_i^n (1 + \kappa\lambda_i). \tag{30}$$

Call an eigenvalue λ_i with $i \geq 2$ *dangerous* if

$$1 + \kappa\lambda_i < 0, \qquad \text{equivalently} \qquad \lambda_i < -\frac{1}{\kappa}. \tag{31}$$

Because n is even, every nondangerous summand $\lambda_i^n(1 + \kappa\lambda_i)$ in (30) is nonnegative.

Case 1: no dangerous eigenvalue exists. Then (30) and (21) give

$$t(C_n, W) + \kappa t(C_{n+1}, U) \geq f_\kappa(\lambda_1) \geq f_\kappa(b) = \rho_n(a).$$

Case 2: a dangerous eigenvalue exists. There is at most one dangerous eigenvalue. Indeed, if two eigenvalues were smaller than $-1/\kappa$, then Lemma 2.5 and Lemma 2.7 would give

$$\frac{1}{4} \geq \sum_{i=2}^{N} \lambda_i^2 > \frac{2}{\kappa^2} > \frac{1}{4},$$

a contradiction. Write the unique dangerous eigenvalue as

$$-\beta, \qquad \beta > \frac{1}{\kappa}.$$

All remaining terms in the sum in (30) are nonnegative. Perron–Frobenius-type domination (15) gives

$$\lambda_1 \geq \beta. \tag{32}$$

Moreover, Lemma 2.7 gives

$$\beta > \frac{1}{\kappa} > b. \tag{33}$$

By (32) and (14),

$$0 < \beta \leq \lambda_1 \leq 1.$$

Since f_κ is strictly convex and minimised at b, it is increasing on $[b, 1]$. Hence, by (32) and (33),

$$f_\kappa(\lambda_1) \geq f_\kappa(\beta).$$

Therefore (30) yields

$$\begin{aligned} t(C_n, W) + \kappa t(C_{n+1}, U) &\geq f_\kappa(\beta) + \beta^n(1 - \kappa\beta) \\ &= (1-\beta)^n + \kappa\beta^{n+1} + \beta^n - \kappa\beta^{n+1} \\ &= (1-\beta)^n + \beta^n. \end{aligned}$$

Since n is even, convexity of $x \mapsto x^n$ gives

$$(1-\beta)^n + \beta^n \geq 2\left(\frac{1}{2}\right)^n = 2^{1-n}. \tag{34}$$

Finally, $a \leq \alpha_n^\star$ and the monotonicity of ρ_n imply

$$\rho_n(a) \leq \rho_n(\alpha_n^\star) = 2^{1-n}.$$

Combining this with (34) proves (29). $\square$

It remains to prove the upper bound. Let W_{cl} be the balanced disjoint union of two cliques:

$$W_{\mathrm{cl}}(x,y) := \begin{cases} 1, & x,y \in [0,1/2] \text{ or } x,y \in (1/2,1], \\ 0, & \text{otherwise.} \end{cases}$$

Since C_n is connected, every homomorphism contributing to $t(C_n, W_{\mathrm{cl}})$ maps all of its vertices into the same half. Hence

$$t(C_n, W_{\mathrm{cl}}) = 2\left(\frac{1}{2}\right)^n = 2^{1-n}. \tag{35}$$

The complement $1 - W_{\mathrm{cl}}$ is the balanced complete bipartite graphon. Since $n+1$ is odd,

$$t(C_{n+1}, 1 - W_{\mathrm{cl}}) = 0. \tag{36}$$

Thus

$$t(C_n, W_{\mathrm{cl}}) + \kappa t(C_{n+1}, 1 - W_{\mathrm{cl}}) = 2^{1-n}. \tag{37}$$

If $a > \alpha_n^\star$, then strict monotonicity of ρ_n gives

$$\rho_n(a) > \rho_n(\alpha_n^\star) = 2^{1-n}.$$

Therefore W_{cl} violates the scaled commonality inequality (19). It follows that

$$\pi(C_n, C_{n+1}) \subseteq (0, \alpha_n^\star].$$

Proposition 2.9 gives the reverse inclusion. Taking $n = 2k$, the number $\alpha_{2k}^\star$ is exactly the number $a_k^\star$ defined in Theorem 1.1, so the theorem follows.

3 Remarks

The proof of Theorem 1.1 has been verified in the Lean 4 proof assistant [6], on top of the mathlib library [5]; the development accompanies this note. The formal statement covers both directions and the existence and uniqueness of $a_k^\star$: the inequality holds for every graphon if and only if $a \le a_k^\star$. The formal proof differs from the one given here in four ways.

First, instead of the dyadic conditional expectations in Lemma 2.1, the formal proof approximates W directly by a step graphon in the L^1 norm; both cycle densities are Lipschitz in that norm, so the inequality transfers.

Second, the interlacing inequalities (11) are proved by an intersection argument instead of the Courant–Fischer principle: two spans of eigenvectors whose dimensions add up to more than N contain a common unit vector, and evaluating the quadratic forms of A and $A+P$ at that vector gives the inequality.

Third, the rearrangement in Lemma 2.3 is not obtained by sorting. Since the α_i are already ordered, rearranging $(\alpha_1, \ldots, \alpha_{N-1}, \alpha_N + 1)$ only moves the entry $\alpha_N + 1$ to its correct position, and the formal proof writes this down explicitly, with the partial sums in closed form. Karamata's inequality (9) is proved in a form that requires only one of the two sequences to be monotone, so the eigenvalues μ_i of $A+P$ are not sorted either.

Fourth, the derivatives of f_κ are not computed. The paper locates the minimum of f_κ at b through f_κ' and f_κ''; the formal proof instead bounds $(1-x)^n$ and κx^{n+1} from below by their supporting lines at $x = b$, both instances of Bernoulli's inequality, and the sum of the two lines is the constant $\rho_n(a)$ because $\kappa(n+1)b^n = na^{n-1}$.

References

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